



STRUCTURAL DESIGN OF A FOIL USING GEOMETRIC PROGRAMMING: MODELING AND SOLUTION

Project Report for ENEL 667 Optimization for Engineers

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1 INTRODUCTION

The purpose of this project is to learn about a specific topic on optimization. I chose to learn about geometric programming, and for an application I selected an aspect of structural design of a foil¹: minimization of the volume of material making up the foil while meeting certain requirements, for example on maximum deflection and stress.

The optimization problem was first modeled. Then, a relaxed form of the model was used to facilitate gaining some experience with a basic solver. Two versions of the same relaxed model, one in basic nonlinear form and one in GP convex form, were solved to compare general solver capability for different formulations of the model. Then one scenario of the main foil problem, using the detailed model in GP convex form, was solved using both the basic solver and a more sophisticated one to compare solver algorithm efficiency. Three additional scenarios of interest of the foil model, with more variables, were solved using the more sophisticated solver.

2 PROBLEM DESCRIPTION

The challenge is to minimize the volume of material making up a generic foil. The foil is modelled as a cantilever, with a net distributed load caused by dynamic lift that is applied to the entire foil profile. There are three decision variables: the chord (b) and height (h) of the foil profile, and the skin thickness (t), all of which vary along the span. The main constraints are maximum stress at any point on the foil and maximum deflection at the tip of the foil. Also, it is desired to have a planform area of the entire foil close to but not exceeding a certain value. This planform area establishes the maximum amount of lift that the foil could provide under unspecified conditions of fluid flow. The foil is monocoque in design with a hollow profile, except for a single internal spar element providing additional structural support. It is assumed that further structural elements, including ribs and stringers, would be required for structural integrity of the foil when subjected to load, especially to prevent buckling; however, consideration of these kinds of elements is beyond the scope of this analysis.

3 MODEL DEVELOPMENT

The mathematical formulation of the optimization problem is described in this section. Details follow the summary formulation table shown next

¹ The term foil is used to mean any object that when placed in a moving fluid at a suitable angle of attack, a lift force is produced in a direction perpendicular to the incoming flow of fluid. The lift force is generally greater than the drag force produced, with drag force being parallel to and in the direction of fluid flow. Airfoils (aircraft wings) and hydrofoils are examples of foils.

Table 1. Optimization problem in standard form.

Minimize: volume	$\sum A_{xi}(b_i, h_i, t_i) l$	$i = 1, \dots, n$	(1)
s.t.: max deflection of tip	$y_1(b_i, h_i, t_i) \leq y_{max}$	$i = 1, \dots, n$	(2)
max stress	$\sigma_{max i}(b_i, h_i, t_i) \leq \sigma_{max}$	$i = 1, \dots, n$	(3)
max planform area	$\sum A_i(b_i) \leq A_{max}$	$i = 1, \dots, n$	(4)
monotonic increasing chord	$b_i \leq b_{i+1}$	$i = 1, \dots, n - 1$	(5)
bound on height	$h_{min} \leq h_i \leq h_{max}$	$i = 1, \dots, n$	(6)
bound on skin thickness	$t_{min} \leq t_i \leq t_{max}$	$i = 1, \dots, n$	(7)
ratio of height to chord	$\frac{h_i}{b_i} = r_{des i}$	$i = 1, \dots, n$	(8)
desired skin thickness mid-span	$t_i = t_{des}$	$i = 0.4n, \dots, 0.6n$	(9)

Geometric programming was studied very early in the project as an optimization technique and to inspire problem ideas. It was during this study that I decided on a foil design application; therefore, it was known early on that geometric programming was a strong candidate solution technique for this problem.

A geometric program (GP) is an optimization problem formulation that has the objective and constraints made up of monomials and posynomials [1] [2]. In this form, the optimization problem is not convex. However, after a change of variables and a transformation of the objective and constraints, it can be converted to a convex optimization problem. This fact was used to guide development of the model formulation.

A symmetrical National Advisory Committee for Aeronautics (NACA) airfoil profile was used for the cross-sectional shape of the foil. Although the precise shape of the foil is not important for this study, this profile was used to provide a visual reference and starting point for certain formulas described below. A typical symmetric NACA profile is as follows

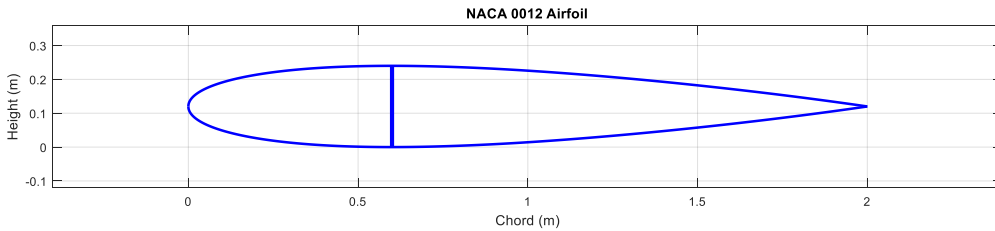


Figure 1: NACA airfoil and spar cross-section; chord of 2 m; arbitrary height to chord ratio of 12% and no camber.

The first step in the modeling process was to establish the variables of the problem. The approach taken was to divide the foil into segments along the span, in such a way that each segment volume would consist of an area cross section, such as shown in Fig. 1, together with a constant width l of each segment. The variables are then b , h , and t for the chord, height, and the skin thickness of the foil segments, respectively. There are n segments, therefore there are a total of $3n$ variables. The result is a kind of finite element model of the foil. By increasing n , the segment width l decreases, thereby resulting in a finer “mesh”.

A distributed load due to aerodynamic lift on the foil is the only source of loading considered. This load generates a bending moment along the span of the foil. Shear force due to the distributed load also contributes to stress in the foil; however, the significant majority of stress results from bending and thus the contribution from shear forces is handled within the safety factor applied to the yield stress of the material. Lastly, the aerodynamic drag force parallel to the chord of the wing is assumed to contribute a negligible amount to the stress in the foil skin due to the high moment of inertia about the axis perpendicular to the span of the foil.

The bending moment, which varies as a function of distance from the foil tip, was derived using a distributed load that has two changeable components, w_t and w_r , as shown below

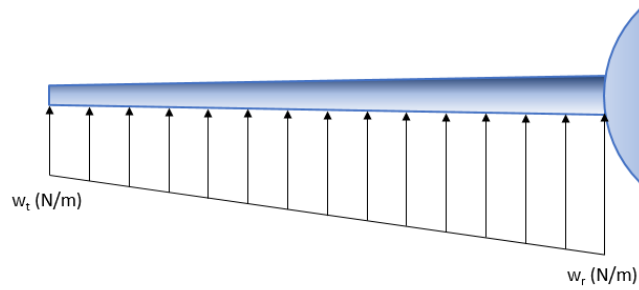


Figure 2: Distributed load due to lift acting on foil; front view.

The bending moment is given by

$$M_i = \frac{(w_r - w_t) l^3 i^3}{6 L} - \frac{w_t l^2 i^2}{2} \quad (10)$$

Where,

M_i = bending moment for segment i , Nm

w_r = foil root load (variable), N/m

w_t = foil tip load (variable), N/m

l = segment thickness (constant), m

i = segment #, measured from tip to root

L = width of foil (tip to root), m

The maximum stress (due to bending) in any segment of the foil is given by

$$\sigma_{\max i} = \frac{M_i h_i}{2 I_i} \quad (11)$$

Where,

$\sigma_{\max i}$ = maximum stress in segment i , Pa

h_i = height of foil cross-section for segment i , m

I_i = moment of inertia for segment i , m⁴

The cross-sectional area of a segment of the foil is

$$A_{Xi} = \left[\frac{1.2}{r_i} + 1.2 + a \right] h_i t_i \quad (12)$$

Where,

A_{Xi} = segment area, m²

r_i = ratio of height to chord for segment i , d-less

a = ratio of spar thickness to skin thickness, d-less

t_i = foil skin thickness for segment i , m

The deflection of a cantilever at the tip requires knowledge of the geometries of all segments in the foil. For example, deflection of any given segment requires knowledge of both the slope and deflection of the adjacent segment in the direction of the root of the foil. In this fashion, the deflection constraint is a global constraint that requires a recursive loop calculation. Equations for slope and deflection were derived as

$$\theta_i = \frac{\left[\frac{l^4 (w_r - w_t) [i^4 - (i-1)^4]}{24 L} + \frac{l^3 w_t [i^3 - (i-1)^3]}{6} \right]}{EI_i} + \theta_{i+1} \quad (13)$$

$$y_i = \frac{\left[\frac{l^5 (w_r - w_t) [10 i^3 - 10 i^2 + 5 i - 1]}{120 L} + \frac{l^4 w_t [6 i^2 - 4 i + 1]}{24} \right]}{EI_i} + l \theta_{i+1} + y_{i+1} \quad (14)$$

Where,

θ_i = slope for segment i , radians

E = Modulus of Elasticity, Pa

y_i = displacement for segment i , m

At this point in the formulation process, the moment of inertia of the cross-sectional area of the foil was needed. An approximation given by [3] was used as the starting point. A problem arose during the derivation that threatened to compromise the ability of the problem to be converted to a convex GP. Specifically, a posynomial appeared in the derivation of I_i . The issue with this is that for the deflection constraint, which has multiple terms, I_i appears in the denominator of each of those terms. It was therefore decided to avoid this issue by manually eliminating one variable (b). This was achieved by combining Eq. 8 with all other equations as appropriate. The value of $r_{des i}$ on the RHS of Eq. 8 was set to vary between the values of 0.08 and 0.12 on a linear scale between the tip and root of the foil. After elimination of some higher order terms with minor loss of accuracy, I_i becomes

$$I_i = \left[0.075 \left(\frac{3}{r_i} + 1 \right) + \frac{1}{12} a \right] h_i^3 t_i \quad (15)$$

With Eq. 16, all constraints equations become monomials or posynomials, and the problem can be converted to a GP in convex form. Since the model formulation is convex, any local minimum that is found is indeed the global minimum.

With a change of variables using x_1 and x_2 for h and t , respectively, and transformation of objective and constraints, the optimization problem in GP convex form is

Table 2. Geometric program in convex form.

Minimize: Volume	$\ln \left[\sum \exp[x_{1i} + x_{2i} + \ln(K_{vol i})] \right]$	$i = 1, \dots, n$	(16)
s.t.: max deflection of tip	$\ln \left[\sum \exp[-3x_{1i} - x_{2i} + \ln(K_{y1 i})] \right] \leq 0$	$i = 1, \dots, n$	(17)
max stress	$-2x_{1i} - x_{2i} + \ln(K_{sm i}) \leq 0$	$i = 1, \dots, n$	(18)
max planform area	$\ln \left[\sum \exp[x_{1i} + \ln(K_a i)] \right] \leq 0$	$i = 1, \dots, n$	(19)
monotonic increasing chord	$x_{1i} - x_{1i+1} + \ln(K_r i) \leq 0$	$i = 1, \dots, n - 1$	(20)
bound on height	$\ln(h_{min}) \leq x_{1i} \leq \ln(h_{max})$	$i = 1, \dots, n$	(21)
bound on skin thickness	$\ln(t_{min}) \leq x_{2i} \leq \ln(t_{max})$	$i = 1, \dots, n$	(22)
desired skin thickness mid-span	$x_{2i} - \ln(t_{des}) = 0$	$i = 0.4n, \dots, 0.6n$	(23)

In the above summary, the K values are coefficients used to simplify otherwise complicated expressions containing only constants. These coefficients were calculated using Matlab.

4 PROBLEM SOLUTION

4.1 SIMPLIFIED PRISMATIC CANTILEVER DESIGN SOLUTION

The first step to solving my problem was to find an off-the-shelf optimizer. To quickly get up-and-running with a solver, and not yet knowing what specifications were required, I chose to try the basic Mathworks® Matlab Optimization Toolbox. To start, I used this software together with a relaxed version of my problem. This simplified problem assumes a prismatic cantilever beam instead of a foil, and with fewer constraints. For brevity, the model formulation for the simplified problem is not included here but is very like the one in [1] on (pp. 163-165). One question in my mind was whether the problem could be just as easily solved if the model was left in a non-convex form. To answer this question, two different formulations of the same problem were tested: one using just a nonlinear non-convex model formulation and the other formulation as a GP in convex form. An interior-point algorithm was used with the solver, and successful solution resulted for an arbitrary selection of parameters, constraints, and small to moderate number of variables. A comparison of solution times as a function of number of variables is shown below

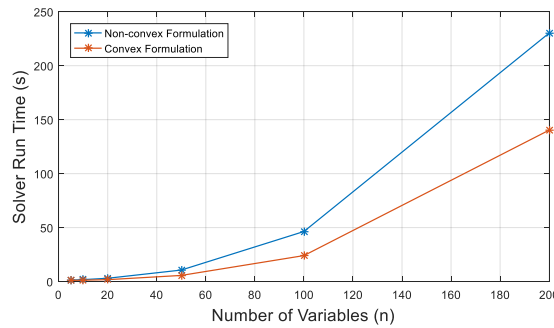


Figure 3: Comparison of solver run time for convex and non-convex formulations of design of cantilever beam using Matlab Optimization Toolbox.

Figure 3 suggests an exponential increase in run time with number of variables using this solver. Since I proposed to solve the foil design problem using about 1000 variables, these results suggest it's prudent to formulate the problem as a GP in convex form, and that a better solver would likely benefit the foil design problem.

To conclude the above exercise, the optimal solution that produced an optimal value of 87.9 is shown below ($n = 50$). Since the same problem was used for both model formulations, the solution was the same for both. The changing character of the width and height profiles at the three to five metre mark is due to constraints becoming active.

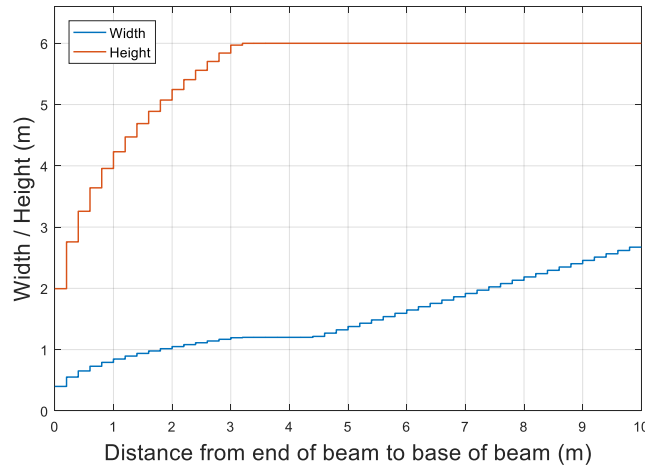


Figure 4: Optimal solution for convex and non-convex formulations of design of cantilever beam using Matlab Optimization Toolbox.

4.2 FOIL DESIGN SOLUTION

Four different scenarios of the foil optimization problem were solved, each with different parameters for the constraints.

At this point the CVX [4] [5] modeling system with the Mosek 8.0 [6] solver was employed. CVX was designed specifically for constructing and solving disciplined convex optimization problems. Mosek 8 is a leading solver package for nonlinear convex optimization and large scale problems. One benefit of using CVX is the built-in GP mode. With this mode, the model is entered in the code in a basic form like that of the entries in Table 1. CVX does the change of variables and transformation to convex form internally and then converts the solution back, thus saving effort. With the Matlab Optimization Toolbox, manual formulation of the model into convex form was required (as per Table 2) prior to entering the model into code.

For the first scenario, considered the base case, both the Matlab Optimization Toolbox and CVX were used to solve the problem and compare results and performance. (For the remaining scenarios, just CVX was used.) Parameters are shown in Table 3, and results are discussed below

Table 3. Geometric, material, and load parameters.

Constraint Parameter	Value	Units	General Parameter	Value	Units
Max deflection (y_1)	0.1	m	Foil span (one side) (L)	4	m
Max stress (σ ; Al 6061-T6) *	127	MPa	Modulus of Elasticity (E; Al 6061-T6)	68.9	MPa
Max planform area (A)	3	m ²	Load distribution tip (w_t)	0	N/m
Min foil height ($h_{\min}$)	0.0005	m	Load distribution root (w_r)	1500	N/m
Max foil height ($h_{\max}$)	0.5	m	Ratio of spar width to skin thickness (a)	5	d-less
Min skin thickness ($t_{\min}$)	0.0001	m			
Max skin thickness ($t_{\max}$)	0.01	m			

* Represents 0.5*yield stress of 255 MPa (safety factor of 2x applied)

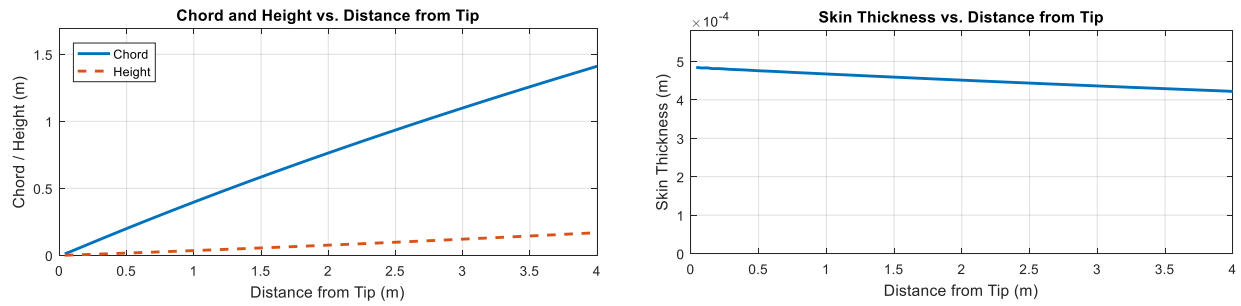


Figure 5: Optimal solution for foil design problem; $n=100$ segments; variables h and t ; max deflection 0.1m.

Both solvers yielded essentially the same solution (Fig. 5). The chord, height, and skin thickness are all parabolic in profile. The total deflection at the foil tip is 0.1 m and the total planform area of the foil is 3 m²; therefore, these constraints are both active as expected. A maximum stress of 69.9 MPa occurs in the segment at the root of the foil; thus, none of the maximum stress constraints are active. Also, none of the bounding constraints on foil height and skin thickness are active. The optimum value is 0.002461 m³, and the material mass is 6.67 kg. Figure 6 compares the solution times for each solver for increasing number of variables.

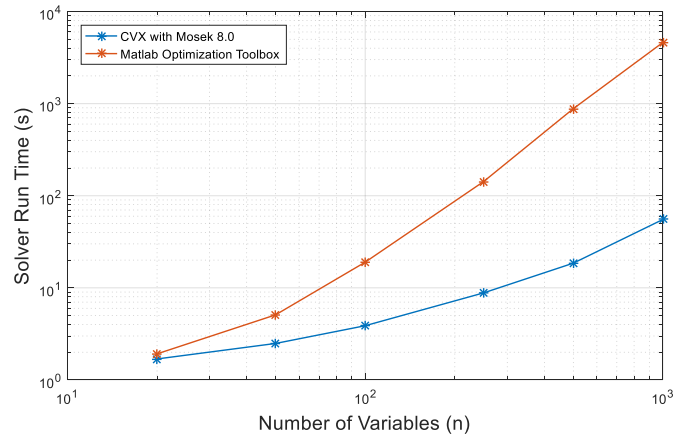


Figure 6: Comparison of solver run time for base case foil design problem.

The Fig. 6 comparison clearly illustrates the benefit of using a solver specifically tailored to convex GP optimization problems for modest number of variables. At $n=500$ and 1000 , the Matlab Optimization Toolbox solution was observed to have numerical instability character near the tip of the foil.

In the next scenario, the maximum deflection (constraint) is relaxed to a value of 0.3m . All other constraint parameters are unchanged. Only CVX was used to solve it.

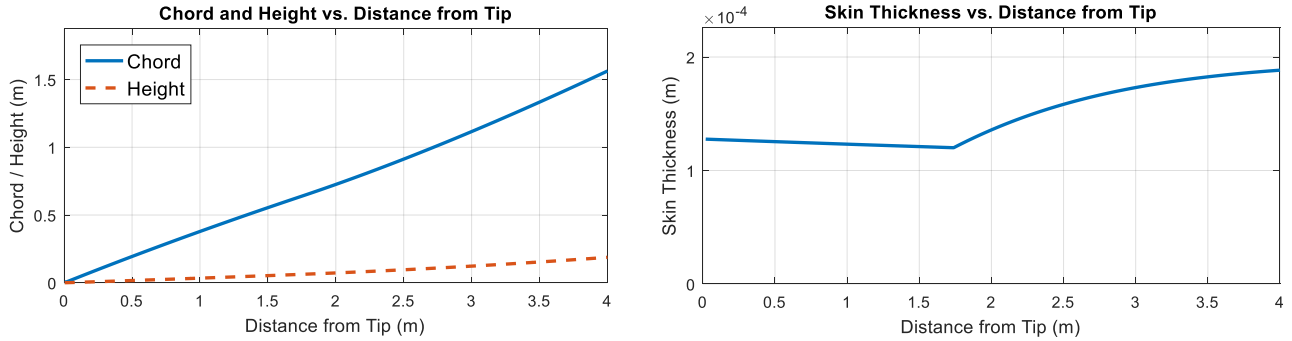


Figure 7: Optimal solution for foil optimization problem using CVX with Mosek 8.0; $n=1000$ segments; max deflection 0.3m .

In this case (Fig. 7), the skin thickness is substantially less than that of the prior case (Fig. 5). In short, the foil doesn't need to be as stiff; therefore, not as much material is required. Also, the chord, height, and skin thickness have different character along the length of the foil. In particular, the skin thickness decreases a minor amount from 0 to 1.7m , but then increases sharply. The reason is that the increased deflection leads to more than half the segments nearest the root reaching maximum stress. Hence, some material is added to the skin thickness to keep these segments from exceeding maximum stress. The maximum stress constraints are active for these segments. As for the prior case, the total deflection and total planform area constraints are active and none of the bounding constraints on foil height and skin thickness are active. The optimum value is 0.000909 m^3 , yielding a material mass of 2.46 kg . The volume is nearly a third as much as for the prior case, illustrating the penalty that added rigidity places on the amount of material required.

The next scenario of the optimization problem restricts the foil to having a constant chord (0.75 m). Since the chord is constant and height is a dependent function of chord, the only true variable is the skin thickness. The distributed load profile was modified to a rectangular profile ($w_t, w_r = 750\text{ N/m}$) to more closely reflect the planform shape of the foil, and the value of r_{des} at the root was set to 0.16 . Two subcases were compared. In the first, there were no other changes to default parameters and constraints. In the second, skin thickness was fixed at 0.002 m for 20% of the span in the middle. The fixed added skin thickness could be a requirement, say, for extra strength for attachments.

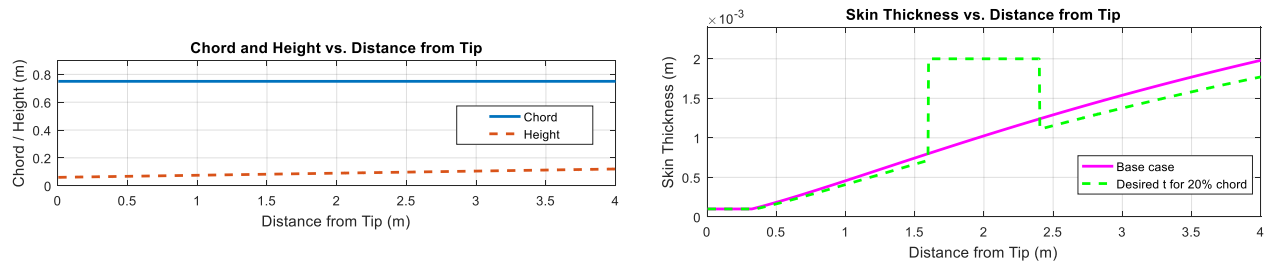


Figure 8: Optimal solutions for foil optimization problem using CVX with Mosek 8.0; $n=1000$ segments; max deflection 0.1m; base case compared to case with fixed skin thickness for 20% of chord.

For the foil having fixed skin thickness for 20% of the span, the skin thickness where it's not fixed is slightly less than for the default case at the same position along the foil (Fig. 8). The reason is that the fixed skin thickness adds rigidity and reduces the deflection, thereby allowing the thickness to be relaxed for the remaining portion of the foil. Note that this is only the case if and where the skin is not fully stressed. If the skin is fully stressed (active constraint), fixing skin thickness for a portion of the span will not allow skin thickness to be reduced anywhere else along the span since that would violate the stress constraint, yielding an overstressed condition. In the above cases, planform area is 3m^2 (active), deflection is 0.1m (active), while the maximum stress and variable bounds constraints are not active. Optimal values were obtained as 0.006112m^3 (16.5646 kg) and 0.006740m^3 (18.2646 kg) for the base and fixed thickness cases, respectively. Note that these are significantly larger values than the last scenario because of the (arbitrarily) increased distributed load profile.

5 CONCLUSIONS

The major outcome of the project was to learn that my foil optimization problem can be solved efficiently by modeling it as a GP in convex form.

To arrive at this conclusion, first considerable effort was put into modelling the problem as a GP. This was successful with minor simplifying assumptions for the moment of inertia of the foil cross-section. The benefit of having the problem modeled as a GP in convex form versus a standard nonlinear form was confirmed using a simplified rectangular beam version of my problem with not many variables. Using a basic solver, Matlab Optimization Toolbox, the convex GP version solved quicker. Next, I learned that a more sophisticated solver, in this case CVX with Mosek 8.0, was ideal for solving various scenarios of interest of the detailed foil problem with many more variables (1000). Matlab Optimization Toolbox was clearly not reliable in this case.

Further work adding more complexity and detail to this problem could include the consideration of aerodynamic forces and additional structural elements, to name a few.

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